

Ethan Cavill

ethancavill@gmail.com | +1 858 518-4580 | +61 420 813 266 | U.S. / Australian Citizen

[linkedin](#) | [github](#) | [portfolio](#)

Experience

Generative AI Startup

Mar 2024 - Present

Lead Software Engineer

San Francisco, CA

- Built and scaled a Laravel + Next.js **consumer product** with **200k+ users signed up**.
- Engineered **real-time chat systems** with **RAG**, memory systems, and TTS.
- Built **GPU-backed image/video pipelines** with ComfyUI workers, cloud uploads, and credit controls.
- Owned **subscriptions, credits, payments/webhooks, moderation, and admin tooling**.
- Led **technical interviewing and engineering hiring**.

Tangible AI

Oct 2023 - Jan 2024

Machine Learning Engineer Intern

San Diego, CA

- Built open-source BERT **search engine with RAG**, achieving **88% improvement** in recall over TF-IDF engine.
- Presented information retrieval findings to **30+ attendees at San Diego Python User Group**.
- Proofread and **published edits to 'Natural Language Processing in Action'**.

Adelaide MRI

Jul 2021 - Oct 2023

Full Stack Software Engineer

Adelaide, SA

- Developed patented radiology reporting software with Angular, Flask, and ML; **boosting output 2-4x**.
- Automated invoicing with MediPass payment **API and webhook integration**.
- Wrote **integration/unit tests with PyTest (90%+ coverage)** and E2E tests with Cypress.
- Built patient booking system with React, GCP, and FastAPI; **reduced call burden by 50%**.
- **Managed large codebase version upgrades** and refactored business critical components.
- Led frontend migration from React to Angular, improving maintainability.
- Implemented strict linting/compiler settings; **reduced Sentry errors by 30%**.

Projects

deepworktimer.io | Next.js, FastAPI, PostgreSQL, Redis, Docker, AWS

2024 - Present

- Streak-based productivity system with **50+ weekly active users** through organic growth.
- Containerized deployment with Docker; secure session management with Redis and PostgreSQL.

ethancavill.me | Next.js, Tailwind CSS, Framer Motion, MDX, Vercel

2020

- Developed a portfolio website with React and Markdown to showcase projects, services, and written insights.

devarity.com | PHP, MySQL, Bootstrap, JQuery

2014 - 2015

- Developed an e-commerce platform **launching with 50+ titles** and partnerships with four indie publishing studios.

Skills

Languages	Python, JavaScript, TypeScript, Golang (Go), PHP, SQL, HTML/CSS, Bash
Technologies	React, Angular, Next.js, Redux, Zustand, RxJS, Docker, Linux, Flask, FastAPI, Django, Laravel, PostgreSQL, MongoDB, Redis, Celery, RabbitMQ, PyTorch, ComfyUI, PyTest, Cypress, Sentry, Selenium, Grafana, GCP, AWS
Concepts	Full-Stack Development, Frontend Development, Backend Development, OOP, Functional Programming, Systems Design, Reactive Programming, State Management, Technical Writing, Agile, Scrum, Algorithms and Data Structures, Machine Learning, Data Science, Distributed Systems, REST, GraphQL, Integration Testing, Unit Testing, End-to-End Testing, Version Control, Git, CI/CD, DevOps, Observability

Education

Bachelor of Mathematical Sciences

Mar 2017 - Nov 2018

University of Adelaide

Adelaide, SA

- Coursework in linear algebra, calculus, and algorithms foundational to CS and ML.
- Left to pursue software engineering career.